HEC

HYDRAULIC EXTERNAL CUTTER

FISHING

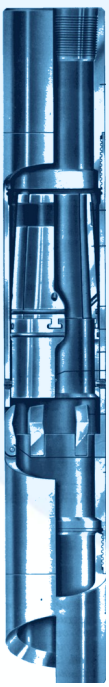
GENERAL DESCRIPTION

HYDRAULICALLY ACTUATED EXTERNAL CUTTER THAT CUTS AND RECOVERS TUBING OR DRILL PIPE IN A SINGLE TRIP. RUN ON A WASHOVER STRING, IT CUTS STUCK PIPE FROM THE OUTSIDE: THE KNIVES ARE FED BY PUMP PRESSURE, GIVING FAST, SMOOTH AND CLEAN CUTS UNDER SENSITIVE OPERATOR CONTROL. IT CUTS ALL TYPES OF DRILL PIPE AND TUBING, REGARDLESS OF TOOL-JOINT STYLE.

TECHNICAL CHARACTERISTICS

Cutter ref.	Pipe to cut	Cutter O.D.	Cutter I.D.	Max. size passed over*	Min. hole to run in	Wt (lb)
70086	1.315 - 2-1/16	3-3/4	2-7/8	2-3/4	4	55
34415	1.315 - 2-3/8	4-1/16	3-3/16	3-1/16	4-5/16	60
70209	1.900 - 2-7/8	4-7/16	3-5/8	3-7/16	4-11/16	73
41727	1.900 - 2-7/8	4-11/16	3-7/8	3-3/4	4-15/16	85
34551	2-1/16 - 4	5-5/8	4-5/16	4-3/16	5-7/8	95
51148	2-3/8 - 4	6-1/16	4-7/8	4-13/16	6-1/4	105
35957	3-1/2 - 5	8	6-1/2	6-3/8	8-1/2	230

* Maximum pipe O.D. the cutter will pass over, dressed with the largest piston segment. Dimensions in inches, weights in lb (approximate).



DESIGN FEATURES

- **Cuts and recovers:** tubing or drill pipe in one trip, inside a washover string.
- **Hydraulic feed:** pump pressure drives the knives; sensitive control of the cut.
- **Run-in protection:** shear pins hold the piston until the cut starts (about 145 psi).
- **Interchangeable piston segments:** one cutter covers several pipe sizes; threaded-edge segments pass collars.
- **Cool, flushed cut:** each segment has a fluid passage for flushing and cooling.
- **Hardened knives:** tempered tool steel, replaceable in the field.
- **Guide options:** standard cut-lip, extra long wall hook or mill-toothed.

OPERATING NOTES

- **Before the run:** wash over the stuck pipe to centralise it at the cut point.
- **Avoid surging:** pressure surges can break a knife tip.
- **After the cut:** torque drops and the weight indicator rises; raise the string slowly to confirm.

PERFORMANCE PARAMETERS

Rotary speed while cutting	15 - 25 rpm
Shear pins	1 pin: about 377 lb; 2 pins: 754 lb
Proving the cut	Torque drop, weight rise
Cut position	1 joint above washover shoe

PULL CAPACITY AND HYDRAULICS

Cutter ref.	Max. load (lb)	psi	gpm
70086	2,530 - 7,910	20 - 40	120 - 220
34415	1,860 - 3,340	20 - 45	120 - 220
70209	1,550 - 3,570	10 - 25	120 - 145
41727	1,128 - 1,780	10 - 30	100 - 200
34551	6,100 - 16,800	15 - 55	180 - 230
51148	2,350 - 4,800	15 - 40	125 - 200
35957	5,160 - 9,040	10 - 20	140 - 180

Approximate values, depending on pipe size. Static loads: if shock loaded, reduce by 50%.

HOW TO ORDER

Please specify: 1. cutter ref. and size of pipe to cut 2. piston segment size 3. connection (top and bottom) 4. guide type (standard, wall hook or mill-toothed) 5. recommended spares: knives, knife pins, shear pins, piston and rubber rings.